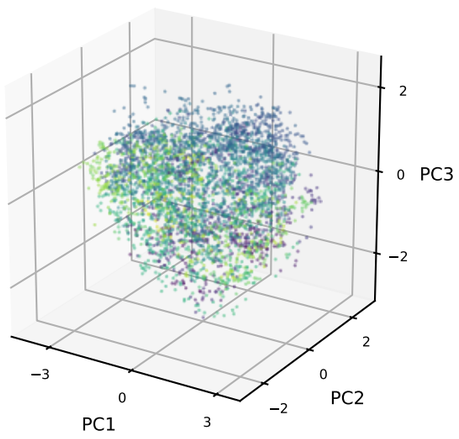
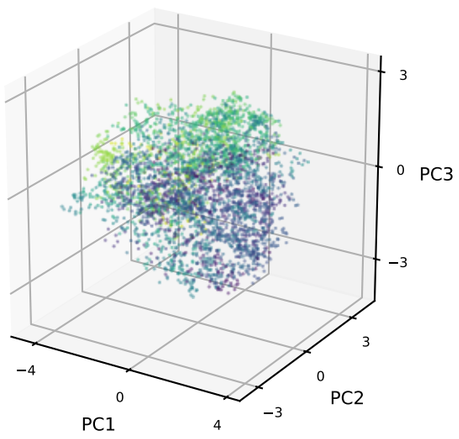
Frozen 1B LSTMs · matched replay across all eight seed pairs
Same PCA coordinates, colored by remembered pickup location; 1 pre-discovery sample in gray

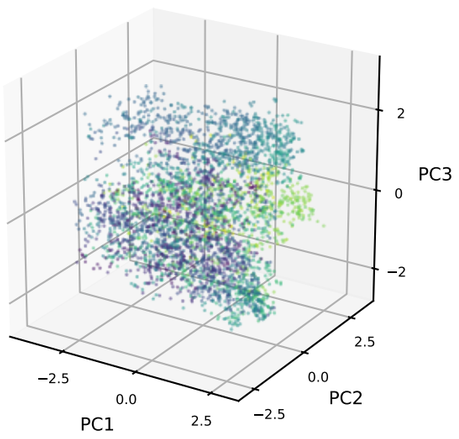
Seed 1 · Grid only
PC1–3: 24.9% of variance



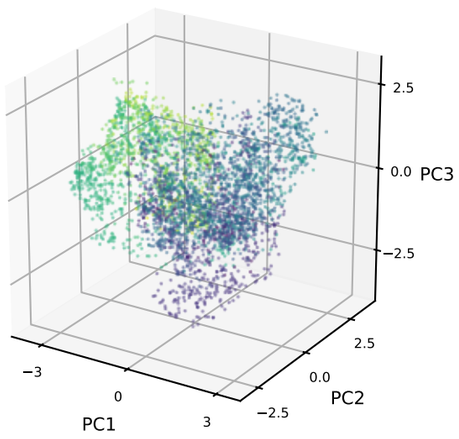
Seed 1 · Grid + 33% PGC
PC1–3: 26.5% of variance



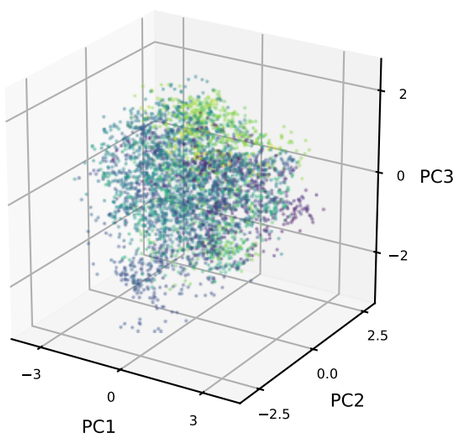
Seed 2 · Grid only
PC1–3: 24.3% of variance



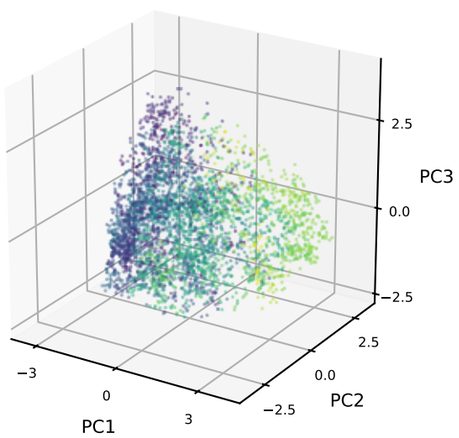
Seed 2 · Grid + 33% PGC
PC1–3: 27.1% of variance



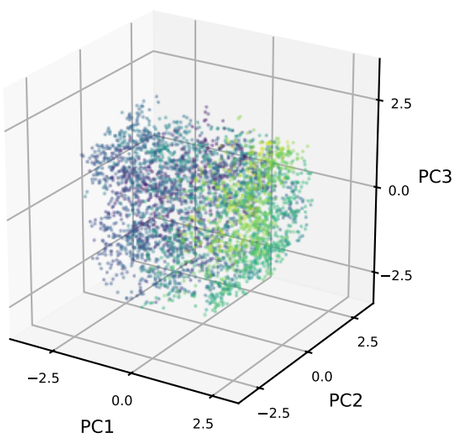
Seed 3 · Grid only
PC1–3: 22.6% of variance



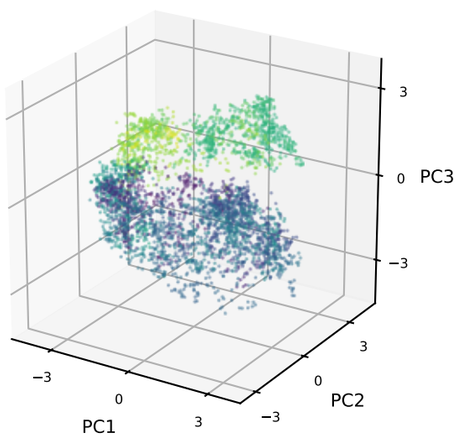
Seed 3 · Grid + 33% PGC
PC1–3: 29.5% of variance

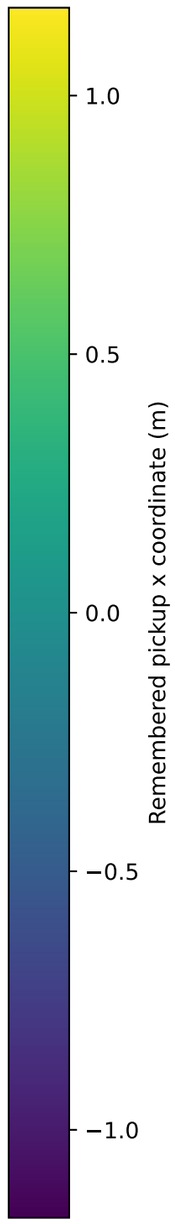


Seed 4 · Grid only
PC1–3: 25.2% of variance

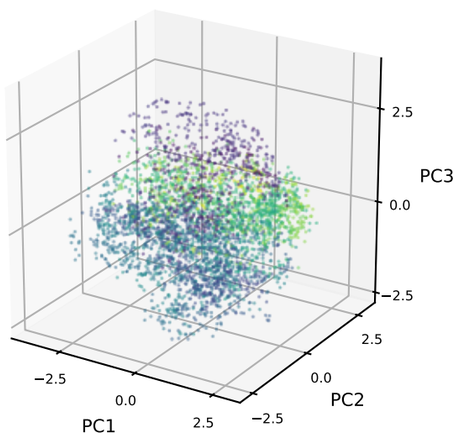


Seed 4 · Grid + 33% PGC
PC1–3: 30.9% of variance

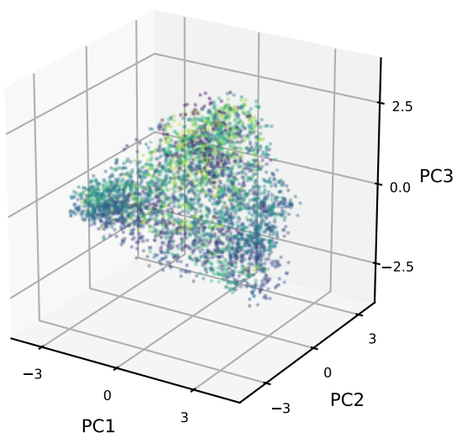




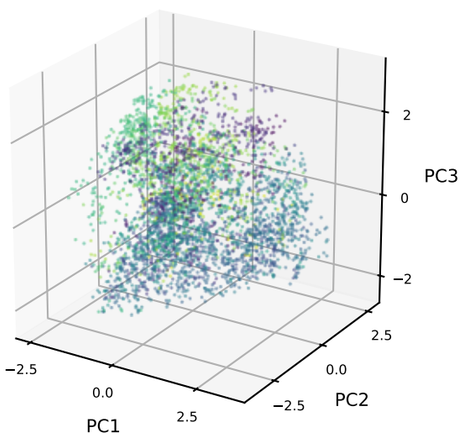
Seed 5 · Grid only
PC1–3: 23.6% of variance



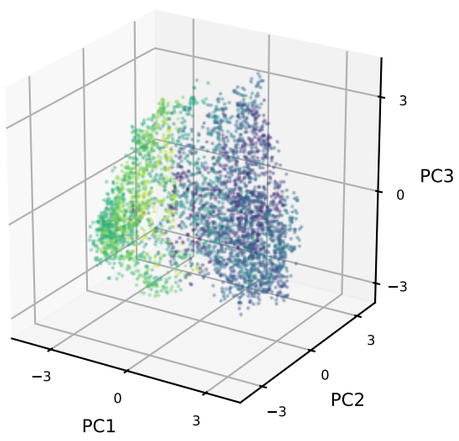
Seed 5 · Grid + 33% PGC
PC1–3: 30.3% of variance



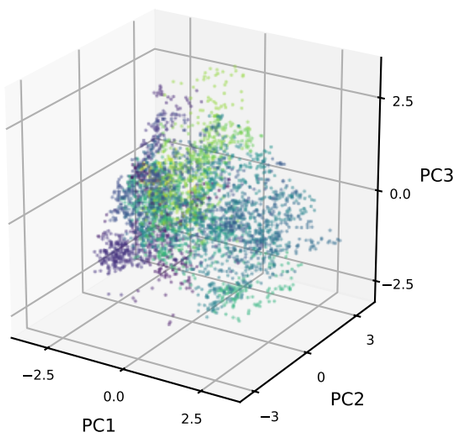
Seed 6 · Grid only
PC1–3: 24.5% of variance



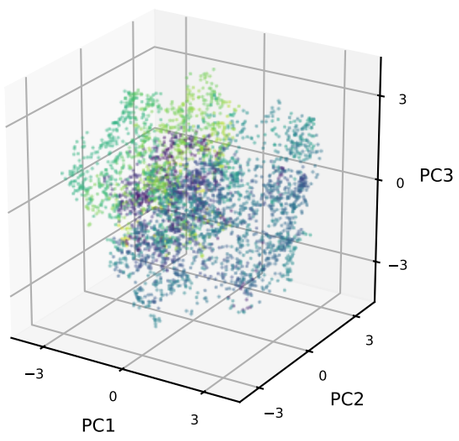
Seed 6 · Grid + 33% PGC
PC1–3: 33.1% of variance



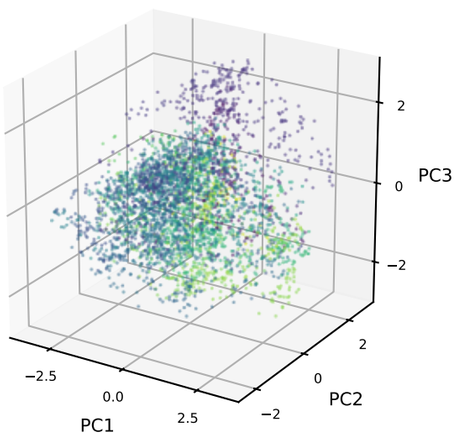
Seed 7 · Grid only
PC1–3: 21.2% of variance



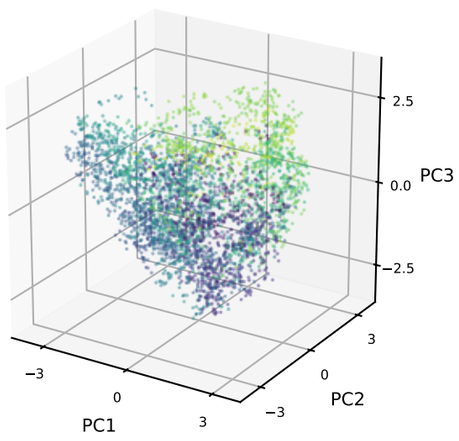
Seed 7 · Grid + 33% PGC
PC1–3: 28.3% of variance



Seed 8 · Grid only
PC1–3: 24.2% of variance



Seed 8 · Grid + 33% PGC
PC1–3: 29.5% of variance



PCA fitted separately to each model; axes are not aligned. Pooling goals and directions does not establish or exclude a torus.